

An Eulerian Note on the Accelerated Collatz Map

Abstract

We discuss the accelerated odd Collatz map

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}} \quad (n \text{ odd}).$$

The note keeps together three elementary structures which are often treated separately: finite products along odd orbits, circular phases obtained from fractional logarithms, and the residue obstruction modulo 3. The purpose is to present an Eulerian unified observation: the accelerated map may be read at once as an exact multiplicative balance, a motion on the unit circle, and a one-way passage through residue classes. In this language a non-principal obstruction, if visible in the representation, would have to preserve a finite product balance, a non-principal circular mode, and compatibility with the many-to-one structure of the map on the two non-zero residue classes modulo 3. The result is not presented as a proof of the Collatz conjecture; it is a compact classical formulation of what a self-sustaining non-principal branch would have to carry.

Keywords. Collatz problem; accelerated Collatz map; finite products; circular sums; invariant branches.

1 Introduction

The Collatz problem is famous partly because its statement is elementary while its known approaches quickly separate into several languages: congruences, trees of preimages, probabilistic heuristics, 2-adic valuations, and estimates for long orbit segments. General background may be found in Lagarias' survey and edited volume [2, 3], and in Wirsching's dynamical treatment of the $3n+1$ function [6]. This short note records a classical way of keeping some of these languages together.

The problem is usually stated in terms of the iteration

$$n \mapsto \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ 3n+1, & n \equiv 1 \pmod{2}. \end{cases}$$

It is often convenient to pass to the accelerated odd map

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}}, \quad n \text{ odd}.$$

The note does not begin with a probabilistic model. Instead it keeps track of the exact finite product along an odd orbit, the corresponding circular phase on the unit circle, and a simple but rigid residue obstruction modulo 3. These are not introduced as independent devices. They are three readings of the same finite motion: size, phase, and residue.

The method is Eulerian in spirit. It uses finite products, explicit fractional formulae, circular sums, and invariant sets, but it tries to postpone the modern habit of separating these into unrelated techniques. The guiding observation is that a divergent or non-principal invariant obstruction, if it exists and is visible in this representation, must survive all three readings at once.

We shall not phrase the final conclusion as a completed proof of the Collatz conjecture. The aim is narrower: to display a classical unified viewpoint and to extract from it the branch condition that a non-principal obstruction would have to carry. This is complementary in spirit to density and stopping-time results such as those of Terras, Everett, and Tao [5, 1, 4], but the present argument is organized around finite identities and circular sums rather than an almost-all conclusion.

2 An Eulerian Point of View

The accelerated odd map has a useful feature: every step removes powers of 2 after applying $3n + 1$. This simple operation can be read in three classical ways.

First, it is a finite product problem. Along an odd orbit one compares the enlargement by $3 + 1/n_j$ with the exact powers 2^{a_j} removed at each step. This keeps the arithmetic balance visible without replacing it by a statistical average.

Second, it is a circular problem. If one looks at the fractional parts of $\log_2 n_j$, then division by powers of 2 disappears on the circle, while the factors $3 + 1/n_j$ become angular displacements. The orbit is therefore not only a sequence of integers but also a sequence of phases.

Third, it is a residue problem. The accelerated image never lies in the class 0 (mod 3). Thus the forward dynamics leaves a persistent empty class and forces any non-principal invariant object into the two remaining classes, where the map has a many-to-one preimage structure.

The point of the note is that these three facts should be read together. A putative exceptional branch is not free to satisfy only one of them. It must carry an exact product balance, keep a detectable circular signature, and remain compatible with the one-way modulo 3 structure. The following sections spell out this observation in finite terms.

3 The Accelerated Odd Map and Finite Products

Let n_0 be odd and define

$$n_{j+1} = T(n_j) = \frac{3n_j + 1}{2^{a_j}}, \quad a_j = v_2(3n_j + 1).$$

Then

$$\frac{n_{j+1}}{n_j} = \frac{3 + \frac{1}{n_j}}{2^{a_j}}.$$

Lemma 1 (Finite product identity). *For every finite odd orbit segment $n_0, n_1, \dots, n_m$,*

$$\frac{n_m}{n_0} = \prod_{j=0}^{m-1} \frac{3 + \frac{1}{n_j}}{2^{a_j}}.$$

Proof. Multiplying the identities

$$n_{j+1}/n_j = (3 + 1/n_j)/2^{a_j}$$

from $j = 0$ to $m - 1$ gives the result. □

Thus every finite segment already carries two competing effects: the 3-fold enlargement and the 2-adic removal measured by the a_j 's. No averaging assumption is involved; the balance is an exact finite identity.

4 Explicit Fractional Form

Set

$$A_m = \sum_{j=0}^{m-1} a_j, \quad A_0 = 0.$$

Lemma 2 (Explicit finite orbit formula). *There is an integer $B_m \geq 0$ such that*

$$n_m = \frac{3^m n_0 + B_m}{2^{A_m}}.$$

It is determined by

$$B_0 = 0, \quad B_{m+1} = 3B_m + 2^{A_m},$$

and hence

$$B_m = \sum_{i=0}^{m-1} 3^{m-1-i} 2^{A_i}.$$

Proof. The statement is immediate for $m = 0$. If

$$n_m = \frac{3^m n_0 + B_m}{2^{A_m}},$$

then

$$n_{m+1} = \frac{3n_m + 1}{2^{a_m}} = \frac{3^{m+1} n_0 + 3B_m + 2^{A_m}}{2^{A_m + a_m}}.$$

Since $A_{m+1} = A_m + a_m$, the recurrence follows. Iterating the recurrence gives the displayed sum. $\square$

Corollary 1 (Periodic balance). *If $n_m = n_0$ for a finite odd cycle of length m , then*

$$(2^{A_m} - 3^m)n_0 = B_m.$$

In particular,

$$2^{A_m} > 3^m.$$

Proof. Substitute $n_m = n_0$ into the explicit formula. Since $B_m > 0$ for $m \geq 1$ and $n_0 > 0$, the inequality follows. $\square$

This is the algebraic form of the balance any non-principal cycle would have to satisfy. It is the first place where a possible exceptional branch must carry a finite product constraint, not merely a qualitative growth pattern.

5 Circular Phase Representation

Let

$$\theta_j = \{\log_2 n_j\}, \quad z_j = e^{2\pi i \theta_j} \in \mathbb{T}.$$

Division by 2^{a_j} changes only the integer part of $\log_2(3n_j + 1)$. Hence the circular displacement from n_j to n_{j+1} is determined by

$$\log_2 \left(3 + \frac{1}{n_j} \right).$$

Indeed,

$$\log_2 n_{j+1} = \log_2 n_j + \log_2 \left(3 + \frac{1}{n_j} \right) - a_j.$$

Summing gives

$$\log_2 \frac{n_m}{n_0} = \sum_{j=0}^{m-1} \log_2 \left(3 + \frac{1}{n_j} \right) - A_m.$$

The main scale comparison is therefore

$$A_m \quad \text{versus} \quad m \log_2 3.$$

Equivalently, if the odd segment is written as the symbolic word

$$1 0^{a_0} 1 0^{a_1} \dots 1 0^{a_{m-1}},$$

then its effective logarithmic displacement is

$$m \log 3 - A_m \log 2.$$

Negative displacement corresponds to

$$\frac{A_m}{m} > \log_2 3.$$

6 Finite Circular Sums

For $k \geq 1$, define the finite circular sum

$$S_k(n_0; m) = \frac{1}{m} \sum_{j=0}^{m-1} e^{2\pi i k \theta_j}.$$

These sums record whether a finite orbit segment preserves a non-trivial circular mode.

Definition 1 (Non-principal circular persistence). *An infinite branch has non-principal circular persistence if there is a fixed $k \geq 1$ and a sequence $m_r \rightarrow \infty$ such that*

$$\limsup_{r \rightarrow \infty} |S_k(n_0; m_r)| > 0$$

along a non-principal branch.

The definition is intentionally elementary. It does not require an ambient ergodic model; it asks only whether a branch can keep a visible circular mode from being averaged away in its finite trigonometric sums.

Remark 1. *In modern terminology, the quantities S_k are finite Fourier coefficients of the circular phase sequence. In the present paper they are used only as finite trigonometric sums attached to orbit segments.*

7 The Modulo 3 Obstruction

The accelerated map has a simple one-way residue feature.

Lemma 3 (Modulo 3 exclusion). *For every positive integer n ,*

$$T(n) \not\equiv 0 \pmod{3}.$$

More precisely, if $a(n) = v_2(3n + 1)$, then

$$T(n) \equiv (-1)^{a(n)} \pmod{3}.$$

Proof. Since

$$3n + 1 \equiv 1 \pmod{3}$$

and $2 \equiv -1 \pmod{3}$, division by $2^{a(n)}$ is multiplication by $(-1)^{a(n)}$ modulo 3. Thus

$$T(n) \equiv 2^{-a(n)} \equiv (-1)^{a(n)} \pmod{3},$$

which is either 1 or 2 modulo 3. $\square$

Thus the forward image of the accelerated odd map avoids the class 0 (mod 3). Any forward-invariant non-principal obstruction must live inside the two non-zero residue classes.

Let $E \subseteq \mathbb{N}_{\text{odd}}$ be a nonempty set with

$$T(E) = E.$$

If E is intended to represent a non-principal obstruction, we also require that it does not contain the principal fixed point 1. Then necessarily

$$E \subseteq \{n : n \equiv 1, 2 \pmod{3}\}.$$

The important point is not merely that one residue class is missing. On the remaining two classes the accelerated map has a many-to-one preimage structure. Therefore, in the present representation, a non-principal invariant obstruction cannot be only a set-theoretic object; it must choose a self-sustaining branch through a many-to-one structure while preserving the circular information described above.

8 The Branch Condition

We now isolate the branch condition suggested by the unified reading above.

Definition 2 (Self-sustaining non-principal branch). *A non-principal branch is called self-sustaining if it satisfies the following three conditions:*

- (i) *it remains in the two non-zero residue classes modulo 3;*
- (ii) *it is compatible with $T(E) = E$ for some non-principal invariant set E ;*
- (iii) *it has non-principal circular persistence.*

The first condition is forced by the modulo 3 obstruction. The second is the invariant-set requirement. The third is the circular requirement: in this representation, a branch that survives indefinitely without joining the principal cycle must preserve some non-trivial circular structure if it is to remain visible as a non-principal object.

Proposition 1 (Eulerian branch condition). *Any non-principal obstruction to the accelerated odd Collatz dynamics which is visible through the finite-product and circular-phase representation above determines a self-sustaining non-principal branch in the stated sense.*

Proof. Let E be a non-principal invariant obstruction in this representation. By the modulo 3 exclusion, E is contained in the two non-zero residue classes. Since $T(E) = E$, the obstruction must select branches through the many-to-one preimage structure of T . Visibility in the circular-phase representation means precisely that some non-principal finite trigonometric mode is not averaged away along the selected branch. These are exactly the three stated conditions. $\square$

This proposition is best read as a branch condition rather than as a final theorem. It says that the Eulerian viewpoint has a definite obstruction object: a branch that is simultaneously compatible with the residue exclusion, the many-to-one invariant-set condition, and the persistence of a non-principal circular mode.

9 A Neutral Conclusion

The preceding sections give a classical chain:

finite products $\longrightarrow$ explicit fractions $\longrightarrow$ circular phases
 $\longrightarrow$ finite trigonometric sums $\longrightarrow$ modulo 3 obstruction
 $\longrightarrow$ invariant branches.

No probabilistic hypothesis is needed to state this chain. Each finite orbit segment is represented exactly. The usefulness of the viewpoint is that it refuses to separate size, phase, and residue too early.

The conclusion drawn here is deliberately modest:

In the Eulerian representation above, a visible non-principal obstruction would have to appear as a self-sustaining branch which preserves a circular mode while living inside the two non-zero residue classes modulo 3 and remaining compatible with the many-to-one structure of the accelerated map.

This formulation leaves a compact object for further study. The remaining issue is to understand whether the three requirements above—finite-product balance, circular persistence, and compatibility with the modulo 3 invariant structure—can be simultaneously maintained along a non-principal branch. The point of the present note is simply to make the classical unity of these requirements explicit. Whether this unity is only a useful reorganization, or something sharper, is left to the reader’s judgment.

References

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